

IVAN MAKARENKO

Technical Product Manager · Robotics Engineer · Entrepreneur

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Robotics engineer and entrepreneur with 5+ years building products from NASA asteroid rovers to a profitable software consulting practice. Now completing an M.S. in Technology Management at Columbia University, I combine hands-on technical depth in Python, C++, and embedded systems with product strategy to help teams ship meaningful products. Seeking Summer 2026 PM and TPM internships at robotics, AI, and deep tech companies.

EXPERIENCE

Automation Engineering Intern · *Stantec* | Chandler, AZ Jun – Aug 2024

- Uncovered a 25% error rate in legacy electrical schematics across mining infrastructure projects and drove Revit model updates across 12 projects, reducing cross-team coordination errors by 20% and accelerating design approval cycles.
- Standardized Revit drawing templates for complex electrical systems, cutting rework cycles and enabling faster design reviews across electrical, mechanical, and civil engineering teams.
- Built an Excel-based project tracker that streamlined documentation and improved coordination across a 50-person engineering organization.

New Product Development Intern · *Addictive Desert Designs* | Mesa, AZ Jun – Aug 2023

- Built a vehicle trim database tracking specifications across 50+ truck models, cutting design cycle time by 30% and enabling a 33% increase in quarterly parts output while protecting first-to-market positioning.
- Drove the full product development lifecycle from laser scanning real vehicles through SolidWorks CAD design to robotic assembly and testing, improving manufacturing precision across bumpers, fenders, and chassis components.
- Partnered with OTC engineers to commission two industrial robotic welding systems as part of a 40,000 sq ft facility expansion, ensuring reliable integration for high-volume production.

Founder and IT Systems Consultant · *Freelance Consulting Practice* | Remote Jun 2019 – Present

- Built and scaled a profitable IT consulting practice serving 20+ clients and generating \$20K to \$70K annually through entirely referral-based growth, maintaining zero client churn over 5 years while studying full-time in the U.S.
- Managed end-to-end product lifecycle for every engagement by running discovery interviews to surface root problems, prioritizing features by business impact, shipping custom software and analytics dashboards, and iterating based on measurable outcomes.
- Built reusable systems frameworks across construction, legal, and retail clients, developing core PM instincts around managing ambiguity, scoping work clearly, and measuring outcomes over effort.

SELECTED PROJECTS

Robotic Driving System for NASA Psyche Mission · *ASU Senior Capstone* | Mesa, AZ Jan – Dec 2024

- Led mechanical and electrical subsystem design for a rover built to navigate asteroid 16 Psyche’s surface in near-zero gravity and metallic dust terrain, solving traction constraints by concentrating 70% of rover mass in the wheels and designing independent leg suspension that enabled full 360° turns without reversing.
- Worked directly with NASA mission mentors throughout the design process, pivoting from hydraulic to spring-based actuation based on vacuum requirements and delivering a functional prototype that met all NASA weight, size, speed, and budget specifications.

Wearable Heat Stroke Prevention System · *ASU Embedded Systems Project* | Mesa, AZ Sep – Dec 2023

- Designed and built an Arduino wearable hat with a custom Cadence PCB integrating temperature and humidity sensors to detect heat stroke risk in outdoor workers, featuring an alert system that triggered automatically when risk thresholds were exceeded.
- Developed a full commercialization roadmap including an ML predictive algorithm trained on 1,000+ worker data points, a mobile supervisor dashboard, and OSHA compliance integration, validating strong market demand through field surveys.

Algorithmic Trading System · *Personal Project* | Remote 2024 – Present

- Designed and deployed a fully automated trading system on Google Cloud Platform connecting live exchange APIs across multiple platforms to a custom multi-timeframe algorithmic strategy, executing trades on BTC and ETH perpetual futures with real capital.
- Built the full system architecture independently including real-time market data ingestion, signal generation, risk management logic, and automated order execution, iterating the strategy continuously based on live performance data.

EDUCATION

M.S. in Technology Management · *Columbia University, School of Professional Studies* | New York, NY Expected Dec 2026
Coursework in Product Strategy, Digital Transformation, Agile Leadership, and Innovation Management

B.S. in Engineering, Concentration in Robotics · *Arizona State University, Ira Fulton School of Engineering* | Mesa, AZ Dec 2024
NASA Psyche Mission Capstone · Varsity Swimming Captain, Lake Forest Academy

SKILLS

Product: Product Lifecycle Management · Requirements Gathering · Agile and Scrum · Roadmapping · Cross-functional Leadership · Data-Driven Iteration
Engineering: Python · C/C++ · SQL · MATLAB · Git · Embedded Systems · PCB Design · SPI/I2C · RPA
Design & Tools: SolidWorks · Revit · AutoCAD · Cadence · Tableau · Excel (Advanced) · VS Code · Linux and Windows
Languages: English (Fluent) · Russian (Fluent) · Ukrainian (Fluent)